QF600 Asset Pricing — Lecture 6 Reference Cheatsheet

Stochastic Discount Factor: the consumption–portfolio problem, the Euler equation, $\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$, and the price, excess-return and covariance forms

Built from Lecture 6 slides (Liu Peng, LKCSB, updated 32-slide deck) · October 2026 · with symbol-by-symbol equation breakdowns

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Source: "Lecture slides – Week 6 v2" (Liu Peng, Lee Kong Chian School of Business, updated version, 32 slides). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond*

slides]. △! marks a slide ambiguity or error, with the corrected version.

How to read each entry: a **Concept** paragraph says what the idea is and why it matters; then the key equation; then a shaded where box that names each symbol and says what it represents in that formula; then intuition, worked numbers and traps.

Exam context (Week 1 [S5]): closed book, one double-sided A4 sheet allowed; 40% MCQ and 60% written (concepts plus simple calculations), with **no coding**. From this lecture you need: the two decisions and the budget; what the Euler equation and the allocation rule say; $\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$ with $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$; $P_i = \mathbb{E}[\tilde{M}\tilde{X}_i]$; $\mathbb{E}[\tilde{M}\tilde{R}^e_i] = 0$; $R_f = 1/\mathbb{E}[\tilde{M}]$; the covariance form of the risk premium; and the insurance-vs-procyclical logic. The slides mark the C_0 derivation [S17] **"(optional)"** and the per-asset step [S23] **"(derivation not tested)"**: know those results, not the algebra. In-class quiz checkpoints: Q1–3 [S16], Q4–6 [S22], Q7–9 [S32].

Notation used throughout (slide conventions)

Symbol	Meaning	Type	Notes
W_0	wealth today ("salary + assets")	\$, known	[S3], [S11]
C_0	consumption today (the choice)	\$, chosen	[S3]
$S = W_0 - C_0$	amount saved and invested	\$, known	a level, not a rate
w_i	share of the invested pot in asset i	fraction	$\sum_i w_i = 1$ [S8]
$\tilde{R}_i$	**gross** return on asset i: what \$1 becomes	random	1.05 for a 5% gain [S8]
$\tilde{r}$	net return, $\tilde{R} = 1 + \tilde{r}$	random	only on [S11]
$\tilde{R}_p = \sum_i w_i \tilde{R}_i$	portfolio gross return	random	[S7]
$\tilde{C}_1 = \tilde{W}_1$	date-1 consumption = wealth, all eaten	random	$(W_0 - C_0)\tilde{R}_p$
$U(\cdot)$	utility of consumption	function	$U' > 0$, $U'' < 0$ [S6], [S18]
δ	patience: utility discount factor	number	usually < 1 [S9]
λ	Lagrange multiplier on $\sum w_i = 1$	number	not risk aversion △!
$\tilde{M}$	stochastic discount factor $\delta U'(\tilde{C}_1)/U'(C_0)$	random	IMRS, pricing kernel [S24]
$\tilde{X}_i,\ P_i$	payoff at date 1, price today	\$	$\tilde{R}_i = \tilde{X}_i/P_i$ [S25]
R_f	gross risk-free return	known	$= 1/\mathbb{E}[\tilde{M}]$ [S30]
$\tilde{R}^e_i = \tilde{R}_i - R_f$	excess return of a zero-cost position	random	[S26]
m	certain discount factor $1/(1+r)$	number	[S28]
*	evaluated at the optimum	–	e.g. C_0^*
$p_s,\ M_s,\ q_s$	probability, SDF value, state price of state s	numbers	$q_s = p_s M_s$ *[Beyond slides]*

Notation trap: the tilde marks a random variable [S4]: a known amount times a random return is random, so $\tilde{C}_1$ carries a tilde and S does not. Returns are **gross** everywhere except [S11]. W_0 here is wealth; in QF620 W_t is a Brownian motion.

0. Topic map

```
PREREQUISITES
 W1 expected utility, U' > 0, U'' < 0 (risk aversion)
 W2 portfolio return R_p = Σ w_i R_i, Σ w_i = 1
 W3 CAPM: μ_i − R_f = β_i(μ_M − R_f); negative β ⇒ μ < R_f ("insurance")
 W4 multi-factor models
        |
        ▼
LECTURE 6: THE STOCHASTIC DISCOUNT FACTOR
 ├─ SETUP  [S2–S14]
 |   ├─ Spend or invest: C0 + S = W0                          [S2–S3]
 |   ├─ Uncertainty: C̃1 = (W0 − C0) R̃  (tilde = random)       [S4]
 |   ├─ Diminishing marginal utility: U' > 0, U'' < 0        [S5–S6, S18]
 |   ├─ Budget: Σ w_i = 1,  C̃1 = (W0 − C0) Σ w_i R̃_i          [S7–S8, S14]
 |   ├─ Objective U(C0) + δ E[U(C̃1)];  recap                  [S9–S11]
 |   └─ Two decisions: we choose C0, w; the market decides R̃_i  [S12–S13]
 |
 ├─ OPTIMISE  [S15–S23]
 |   ├─ Problem + Lagrangian                                  [S15]
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```
│   ├─ FOC in C0 → Euler: U'(C0*) = δ E[U'(C̃1*) R̃_p]  (optional)     [S17, S19]
│   ├─ FOC in w_i → E[U'(C̃1*) R̃_i] equal for all i                   [S20]
│   └─ Combine → E[M̃ R̃_i] = 1 for every asset  (derivation not tested) [S21, S23]
│
├─ PRICE  [S24–S31]
│   ├─ M̃ = δ U'(C̃1)/U'(C0): IMRS = SDF = pricing kernel             [S24, S28]
│   ├─ P_i = E[M̃ X̃_i]                                               [S25, S27]
│   ├─ E[M̃ R̃^e_i] = 0                                               [S26]
│   ├─ M̃ large in bad states ⇒ insurance dear, boom assets cheap    [S29]
│   └─ R_f = 1/E[M̃];  E[R̃_i] − R_f = −Cov(M̃, R̃_i)/E[M̃]            [S30–S31]
│
└─ CONSEQUENCES  [Beyond slides]
    ├─ state prices q_s = p_s M_s; risk-neutral probabilities
    ├─ Hansen–Jagannathan bound; equity premium puzzle
    └─ M̃ = a − b R̃_M ⇒ CAPM; linear in factors ⇒ W4 models
       │
       ▼
NEXT (L7): many periods, Bellman equation, M_{t,t+i} = δ^i U'(C_{t+i})/U'(C_t)
```

Dependency logic:

- **Setup** supplies two ingredients: a budget linking today to tomorrow, and a concave U that makes a dollar more valuable when consumption is low.
- **Optimise**: at the best plan the investor can't gain by moving a dollar between today and tomorrow (Euler) or between assets (allocation). Together these say every asset satisfies $\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$.
- **Price**: $\tilde{M}$ converts each state's dollars into today's dollars. Every pricing statement (price, excess return, risk-free rate, risk premium) is the same equation rearranged.
- Every model from W3–W5 is a choice of $\tilde{M}$; Lecture 7 chains one-period $\tilde{M}$'s over many dates.

1 Concept entries

1.1 Spend or invest: the one trade-off [S2–S3]

Concept. Asset prices start from one personal decision: how much of today's wealth to consume and how much to set aside. Every dollar invested is a dollar not enjoyed today, so an asset is worth buying only if what it pays later is worth at least the consumption given up now. Prices will be pinned down exactly where the investor is indifferent between eating and investing one more dollar.

$$C_0 + S = W_0, \qquad S = W_0 - C_0.$$

where

W_0	wealth today ("salary + assets" [S11]).
C_0	"SPEND — consume it now"; the investor's choice.
S	"INVEST — set it aside to grow"; a dollar *level*, not a saving rate (S/W_0 would be the rate).

- "The one trade-off in the whole model" [S3]: enjoyment now vs more money later.
- There is no labour income at date 1: tomorrow's consumption comes only from savings. *[Beyond slides]* Adding income Y_1 gives $\tilde{C}_1 = S\tilde{R}_p + Y_1$; the definition of $\tilde{M}$ does not change.

1.2 Uncertainty: a known amount times a random return [S4, S8, S11]

Concept. The investor controls how much to invest, not what it earns. The return is random — one value per future *state of the world* (recession, normal, boom) — so tomorrow's consumption is random too. Think of every random quantity as a list of numbers, one per state; that habit makes everything later in the lecture concrete.

$$\tilde{C}_1 = (W_0 - C_0)\,\tilde{R}.$$

where

$W_0 - C_0$	what you invest; chosen with today's dollars, so known.
$\tilde{R}$	**gross** return, "what \$1 becomes" [S8]; e.g. \$1 → \$0.90, \$1.00 or \$1.20 [S4]; random.
$\tilde{C}_1$	money later; known × random = random, hence the tilde.

- Gross vs net: $\tilde{R} = 1 + \tilde{r}$. A 5% gain is $\tilde{R} = 1.05$.
- **Trap:** the pricing rule $\mathbb{E}[\tilde{M}\tilde{R}] = 1$ needs gross returns. With net returns it reads $\mathbb{E}[\tilde{M}\tilde{r}] = 1 - \mathbb{E}[\tilde{M}]$ *[Beyond slides]*.

△! [S11] writes tomorrow's consumption as $(W_0 - C_0) \times (1 + \tilde{r})$, a net return; every other slide uses the gross $\tilde{R}$. They are the same thing, $\tilde{R} = 1 + \tilde{r}$.

1.3 Diminishing marginal utility [S5–S6, S18]

Concept. An extra dollar is worth more when you have little. Utility rises with consumption, but at a falling rate. This single assumption does three jobs: it makes the investor want smooth consumption over time, it makes them dislike risk, and — the job that matters here — it makes the value of a dollar depend on the state in which it arrives. That state-dependence is exactly what the SDF will measure.

$$U'(C) > 0, \qquad U''(C) < 0.$$

where

$U'(C)$	marginal utility: the slope of U at C, i.e. the extra utility from one more dollar. Positive: more is better.
$U''(C) < 0$	the slope falls as C rises (concavity): "a big jump in well-being" when poor, "barely a nudge" when rich [S6]. It is "risk aversion written as calculus" [S18].

- **Why it matters for pricing [S18]:** a unit of consumption is not equally valuable in every state, so future payoffs *cannot be judged by probability alone*: each state must be weighted by U' there.
- **Example** *[Beyond slides]*: log utility, $U'(C) = 1/C$. A dollar when $C = 90$ is worth $\frac{1/90}{1/120} = 1.33$ times a dollar when $C = 120$.
- **Smoothing [S6]:** moving a dollar from a rich date to a poor date raises total utility, so "the smart move is to smooth consumption over time".
- **Trap:** the slope U' prices assets, not the level U.

Common utility functions *[Beyond slides]*:

$U(C)$	$U'(C)$	Relative risk aversion $-CU''/U'$	Implied $\tilde{M}$
$\ln C$	$1/C$	1	$\delta\, C_0/\tilde{C}_1$
$C^{1-\gamma}/(1-\gamma)$ (CRRA)	$C^{-\gamma}$	γ	$\delta\,(\tilde{C}_1/C_0)^{-\gamma}$
$C - \frac{b}{2}C^2$ (quadratic)	$1 - bC$	$bC/(1-bC)$	$\delta(1-b\tilde{C}_1)/(1-bC_0)$, linear in $\tilde{C}_1$

△! [S5] reads "Consuming 1 dollar is worth more you are poor"; the word "when" is missing.

1.4 The budget: one period, two decisions [S7–S8, S13–S14]

Concept. There are two dates and two choices. At date 0 the investor decides (1) how much to consume and (2) how to split the rest across N assets. At date 1 the market delivers the returns and everything is eaten. The split is ours; the returns are the market's: "maximise over what we choose; take expectations over what we do not" [S13].

$$\sum_i w_i = 1, \qquad \tilde{C}_1 = (W_0 - C_0) \sum_i w_i \tilde{R}_i = (W_0 - C_0)\, \tilde{R}_p.$$

where

w_i	share of the *invested pot* in asset i (not a share of W_0); the shares add to 100% [S8].
$\tilde{R}_i$	gross return on asset i; random except for a riskless asset; "the market's draw" [S14].
$\tilde{R}_p$	portfolio return, the weighted average of the asset returns.
$\tilde{C}_1 = \tilde{W}_1$	date-1 wealth, all consumed: no bequest [S7].

- "You pick the weights, the market fills in the returns" [S14].
- *[Beyond slides]* Weights can be any real number (shorting allowed). The slides never say so, but the first-order conditions only hold with equality for every asset if no weight is stuck at a constraint.

1.5 The objective: one number to rank any plan [S9–S10]

Concept. To compare plans, the investor adds today's utility to the discounted expected utility of tomorrow. Two different preferences are packed in: δ measures impatience (time), and the curvature of U measures how much it matters *which state* a dollar lands in (risk). Keep them apart: δ treats every future state the same; U' does not.

$$V(C_0, \tilde{C}_1) = U(C_0) + \delta\, \mathbb{E}[U(\tilde{C}_1)].$$

where

$U(C_0)$	"enjoy consumption now"; known once C_0 is chosen.
$\mathbb{E}[U(\tilde{C}_1)]$	expected utility of what savings buy, averaged over states.
δ	patience: discounts tomorrow's *utility* (not dollars); higher δ = more patient = save more.

- **Recap [S10]:** utility (not wealth) is what counts; tomorrow has two drivers (how we invest, how the market performs); the split sets the total; later counts for less.
- **Time separable** *[Beyond slides]*: today's and tomorrow's utility simply add, with no interaction (no habits, no durables). Later models relax this (habit formation, Epstein–Zin).
- **Trap:** $\mathbb{E}[U(\tilde{C}_1)] \neq U(\mathbb{E}[\tilde{C}_1])$ (Jensen): the investor cares about the spread of outcomes, not just their average.

1.6 The consumption–portfolio problem [S12–S13, S15]

Concept. Choose C_0 and the weights to make V as large as possible, subject to the weights summing to one. The Lagrangian attaches a price λ to that constraint. Differentiating in each choice gives one condition per decision: one for *how much to save* (the Euler equation) and one for *where to put it* (the allocation rule).

$$\max_{C_0,\, w} \; U(C_0) + \delta \mathbb{E}[U(\tilde{C}_1)] \;\text{ s.t. }\; \sum_i w_i = 1; \qquad \mathcal{L} = U(C_0) + \delta \mathbb{E}[U(\tilde{C}_1)] + \lambda\Big(1 - \sum_i w_i\Big).$$

where

$\max_{C_0,w}$	over what we choose: consumption today and the N weights [S13].
$\tilde{C}_1$	with the budget $(W_0 - C_0)\sum_i w_i \tilde{R}_i$ substituted in.
λ	Lagrange multiplier: the shadow value of the constraint (utility gained if the weights could sum to slightly more than 1). Not risk aversion (W2) and not a factor premium (W5).

- $\partial\mathcal{L}/\partial C_0 = 0 \Rightarrow$ Euler equation (§1.7). $\partial\mathcal{L}/\partial w_i = 0 \Rightarrow$ allocation rule (§1.8).

- Assumes an interior optimum and that differentiation can pass through the expectation.

△! [S15] writes $+\lambda(1-\sum_i w_i)$ and [S20] writes $-\lambda(\sum_j w_j - 1)$: identical. [S20] also drops $U(C_0)$ from the weight problem because it does not depend on the split.

1.7 Euler equation: eat or invest? [S17, S19]

Concept. At the optimum, the last dollar is worth the same eaten today or invested and eaten tomorrow. If investing it bought more utility, you would consume less now; if eating it were better, you would consume more. Optimality is indifference at the margin. This "level balance beam" [S19] is the economic content behind every pricing formula that follows.

$$U'(C_0^*) = \delta\,\mathbb{E}\Big[U'(\tilde{C}_1^*)\sum_i w_i^*\tilde{R}_i\Big] = \delta\,\mathbb{E}\big[U'(\tilde{C}_1^*)\,\tilde{R}_p\big].$$

where

$U'(C_0^*)$	utility given up on the last dollar eaten: the marginal cost today.
$\tilde{R}_p$	what that dollar grows into if invested in the optimal portfolio.
$\delta\mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_p]$	discounted expected utility the dollar buys tomorrow: the marginal benefit.
$U'(\tilde{C}_1^*)$ inside $\mathbb{E}$	random and correlated with $\tilde{R}_p$, so it cannot be pulled out: $\mathbb{E}[U'\tilde{R}_p] \neq \mathbb{E}[U']\mathbb{E}[\tilde{R}_p]$.

Derivation [S17] (marked "optional"): substitute the budget into V and differentiate by the chain rule,

$$\frac{dV}{dC_0} = U'(C_0) + \delta\,\mathbb{E}\Big[U'(\tilde{C}_1)\cdot\frac{\partial\tilde{C}_1}{\partial C_0}\Big] = 0, \qquad \frac{\partial\tilde{C}_1}{\partial C_0} = -\sum_i w_i\tilde{R}_i.$$

where

$\partial\tilde{C}_1/\partial C_0$	a dollar eaten is a dollar not invested; it would have grown to $\tilde{R}_p$, hence $-\tilde{R}_p$.
$= 0$	first-order condition; moving the minus term across gives the Euler equation.

- **Balance beam [S19]:** invested side heavier ⇒ consume less now and invest more; today's side heavier ⇒ consume more now. Level = no gain from moving a dollar either way.
- **Worked example** *[Beyond slides]* (log utility): $\frac{1}{C_0} = \delta\mathbb{E}\big[\frac{\tilde{R}_p}{(W_0-C_0)\tilde{R}_p}\big] = \frac{\delta}{W_0-C_0}$, so $C_0^* = \frac{W_0}{1+\delta}$. With $W_0 = 196$, $\delta = 0.96$: $C_0^* = 100$, $S^* = 96$. The log investor's saving does not depend on returns at all (income and substitution effects cancel).
- **Riskless asset only** *[Beyond slides]*: $U'(C_0) = \delta R_f\mathbb{E}[U'(\tilde{C}_1)]$, the textbook consumption-smoothing condition.
- **Traps:** the minus sign from $\partial\tilde{C}_1/\partial C_0$; pulling $U'(\tilde{C}_1)$ out of the expectation.

1.8 Allocation rule: which asset? [S20]

Concept. Shifting a sliver of the pot from asset j to asset i must not change expected utility at the optimum. So every asset must deliver the same *marginal-utility-weighted* return. Assets can have very different expected returns; what they share is their payoff valued state by state with U'.

$$\delta(W_0-C_0)\,\mathbb{E}[U'(\tilde{C}_1)\tilde{R}_i] = \lambda \quad\Longrightarrow\quad \mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_i] = \mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_j] \;\text{ for all } i,j.$$

where

$(W_0-C_0)\tilde{R}_i$	$\partial\tilde{C}_1/\partial w_i$: extra date-1 consumption from one more unit of weight in asset i.
λ	the same number for every asset, which is why all the left-hand sides are equal.
$U'(\tilde{C}_1^*)$	the same random variable in every line: it depends on total consumption, not on asset i. Only the return multiplying it changes [S20].

- **Why expected returns differ:** $\mathbb{E}[U'\tilde{R}_i] = \mathbb{E}[U']\mathbb{E}[\tilde{R}_i] + \mathrm{Cov}(U',\tilde{R}_i)$. Equal totals with different covariances force different $\mathbb{E}[\tilde{R}_i]$.
- *[Beyond slides]* Combining with §1.7, $\lambda = (W_0-C_0)\,U'(C_0^*)$: the utility value of the whole invested pot.
- **Trap:** what is equal across assets is $\mathbb{E}[U'\tilde{R}_i]$, not $\mathbb{E}[\tilde{R}_i]$.

1.9 Every asset obeys $\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$ [S21, S23]

Concept. Combine the two conditions. The portfolio term in the Euler equation is a weighted average of the per-asset terms, and the allocation rule makes those terms all equal, so the Euler equation holds for each asset on its own. Dividing by today's marginal utility (a known number) turns it into one rule that every asset obeys, with one random discount factor shared by all.

$$U'(C_0^*) = \delta\mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_i] \implies \mathbb{E}\Big[\frac{\delta U'(\tilde{C}_1^*)}{U'(C_0^*)}\tilde{R}_i\Big] = 1 \implies \mathbb{E}[\tilde{M}\tilde{R}_i] = 1.$$

where

$\mathbb{E}[U'\tilde{R}_p] = \sum_i w_i\mathbb{E}[U'\tilde{R}_i]$	the portfolio term is a weighted average of asset terms [S21]; all equal, so each equals the average.
$U'(C_0^*)$	known today, so dividing by it is allowed [S23].
$\tilde{M}$	the ratio $\delta U'(\tilde{C}_1^*)/U'(C_0^*)$: one value per state, the same schedule for every asset.
$= 1$	\$1 invested in any asset is worth exactly \$1 today once its payoffs are valued with $\tilde{M}$.

- **Direct check by perturbation** *[Beyond slides]* (log economy of §1.11): eat \$1 less and lose $U'(C_0) = 1/100 = 0.01$ utils. Use it to buy $1/0.32 = 3.125$ recession-only claims; they pay in the recession state (prob. 0.3, $C_1 = 90$), adding $\delta \times 0.3 \times 3.125 \times \frac{1}{90} = 0.01$ utils. Exactly equal: the price makes the investor indifferent.
- **Assumption:** every weight is interior. An asset held at a binding constraint (e.g. no short sales, $w_i = 0$) only satisfies $\mathbb{E}[\tilde{M}\tilde{R}_i] \leq 1$ *[Beyond slides]*.

△! [S21] writes $\mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_i] = \lambda/\delta(W_0 - C_0)$; it means $\lambda/[\delta(W_0 - C_0)]$. [S23] is marked "(derivation not tested)".

1.10 The stochastic discount factor [S24, S28]

Concept. $\tilde{M}$ is an exchange rate between dollars tomorrow and dollars today, one rate per state. Ordinary discounting uses a single number $1/(1+r)$ for every outcome; the SDF replaces it with a schedule that is high where you would be poor (a dollar is precious) and low where you would be rich. δ shrinks all states equally (time); the marginal-utility ratio tilts value toward bad states (risk).

$$\tilde{M} = \delta\frac{U'(\tilde{C}_1)}{U'(C_0)}.$$

where

δ	patience: a dollar later counts for less, whatever happens.
$U'(\tilde{C}_1)$	how badly you want a dollar in that state: large in a bust, small in a boom.
$U'(C_0)$	how badly you want one today; dividing by it converts utils into today's dollars.
$\tilde{M}$	\$ today per \$ delivered in that state; random, one value per state.

- **Three names, one object [S24]:** intertemporal marginal rate of substitution (IMRS), stochastic discount factor, pricing kernel.
- **Why the name [S28]:** *discount* (it moves a future payoff back to today), *stochastic* (a different value in every state, high where you are poor), *factor* (it multiplies the payoff).
- **Certain → random [S28]:** $P = mX$ with $m = 1/(1+r)$ becomes $P = \mathbb{E}[\tilde{M}\tilde{X}]$. *[Beyond slides]* For a sure payoff, $m = \mathbb{E}[\tilde{M}] = 1/R_f$: the ordinary discount factor is the *average* of the stochastic one.
- **Example** *[Beyond slides]* (log, $\delta = 0.96$, $C_0 = 100$; $C_1 = 90, 105, 120$): $\tilde{M} = (1.067,\ 0.914,\ 0.800)$.

△! [S24] says an IMRS is how many units of today's "happiness" a future dollar is worth. $\tilde{M}$ is in today's *dollars*, not utils (utility ÷ utils per dollar); that is why $P = \mathbb{E}[\tilde{M}\tilde{X}]$ comes out as a price.

1.11 The price form: $P = \mathbb{E}[\tilde{M}\tilde{X}]$ [S25, S27]

Concept. A price is the expected payoff with each state's payoff first converted into today's dollars by $\tilde{M}$. This one line prices anything with a payoff — a stock, a bond, an option, a factory — and it is linear: the price of a portfolio is the sum of the prices of its parts. "Many possible payoffs later, one price now" [S27].

$$\tilde{R}_i = \frac{\tilde{X}_i}{P_i}, \qquad \mathbb{E}\Big[\tilde{M}\frac{\tilde{X}_i}{P_i}\Big] = 1 \quad \Longrightarrow \quad P_i = \mathbb{E}[\tilde{M}\tilde{X}_i].$$

where

$\tilde{X}_i$	what the asset hands you at date 1; different in each state.
P_i	price today; known, so it comes straight out of the expectation [S25].
$\tilde{M}\tilde{X}_i$	the payoff's value in today's dollars in one state; average over states to get the price.

Worked example *[Beyond slides]* (checked in §1.16): log utility, $\delta = 0.96$, $C_0 = 100$; recession $C_1 = 90$ ($p = 0.3$), normal 105 (0.4), boom 120 (0.3); $\tilde{M} = (1.0667, 0.9143, 0.8000)$, $\mathbb{E}[\tilde{M}] = 0.9257$, $R_f = 1.0802$.

Asset	Payoff (rec, norm, boom)	Price P	$\mathbb{E}[\tilde{R}]$	Premium $\mathbb{E}[\tilde{R}] - R_f$
Riskless bond	(1, 1, 1)	0.9257	1.0802	0
Insurance (recession only)	(1, 0, 0)	0.3200	0.9375	−14.3%
Stock	(80, 105, 130)	95.20	1.1029	+2.3%
Coin flip, independent of state	0 or 200, prob. $\frac{1}{2}$	92.57	1.0802	0

- Stock: $0.3(1.0667)(80) + 0.4(0.9143)(105) + 0.3(0.8)(130) = 25.6 + 38.4 + 31.2 = 95.2$.
- The coin flip is very risky but independent of $\tilde{M}$, so it is priced like a bond: $100/R_f = 92.57$. **Variance alone is not priced.**
- **Trap:** $P = \mathbb{E}[\tilde{M}\tilde{X}]$, not $\mathbb{E}[\tilde{M}]\mathbb{E}[\tilde{X}]$; the gap is $\mathrm{Cov}(\tilde{M}, \tilde{X})$ (§1.14).

△! [S25] and [S27] say "pound" / £ while the rest of the deck uses dollars; no effect.

1.12 The excess-return form [S26]

Concept. Buy asset i and fund it by borrowing at R_f. The position costs nothing today, so its discounted payoff must be worth zero. This is the form used to test models on return data: it needs no prices, only excess returns.

$$\tilde{R}_i^e = \tilde{R}_i - R_f; \qquad \mathbb{E}[\tilde{M}\tilde{R}_i] = 1 = \mathbb{E}[\tilde{M}R_f] \implies \mathbb{E}[\tilde{M}\tilde{R}_i^e] = 0.$$

where

$\tilde{R}_i^e$	excess return: payoff of the zero-cost position "long i, short the risk-free asset".
$\mathbb{E}[\tilde{M}R_f] = 1$	the pricing rule applied to the risk-free asset.
$= 0$	subtract the two: the price of a zero-cost position is zero ($P = 0$ in $P = \mathbb{E}[\tilde{M}\tilde{X}]$).

- **Trap:** $\mathbb{E}[\tilde{M}\tilde{R}_i^e] = 0$ does *not* mean $\mathbb{E}[\tilde{R}_i^e] = 0$. Expanding the expectation gives the covariance form (§1.14).
- Check (stock): $\tilde{R}^e = (-0.240,\ 0.023,\ 0.285)$, so $\mathbb{E}[\tilde{M}\tilde{R}^e] = -0.0768 + 0.0083 + 0.0685 = 0$ ✓.

1.13 The SDF is largest when a dollar is scarce [S29]

Concept. Because U' falls as consumption rises, $\tilde{M}$ is high in recessions and low in booms. Pricing with $\tilde{M}$ therefore pays the most for payoffs that arrive in bad times. Insurance-like assets (pay in a slump) are dear, so they have low expected returns; boom assets are cheap, so they have high expected returns.

$$\frac{\partial \tilde{M}}{\partial \tilde{C}_1} = \delta \frac{U''(\tilde{C}_1)}{U'(C_0)} < 0 \qquad \text{e.g. } M_{\text{rec}} = 1.067 > M_{\text{norm}} = 0.914 > M_{\text{boom}} = 0.800.$$

where

$U'' < 0$	concavity makes $\tilde{M}$ fall as future consumption rises.
M_s	value today of \$1 delivered in state s, *per unit probability*.

- **"Risky" means covarying with consumption**, not having high variance: the coin flip in §1.11 is volatile but earns R_f.
- In §1.11's table the recession-only claim *loses* 6.25% on average ($\mathbb{E}[\tilde{R}] = 0.9375$) and is still rationally held: it is insurance.

△! [S29] calls $\tilde{M}$ "the value today of one dollar delivered in that future state". Strictly that is per unit probability; the price of \$1 paid only in state s is the state price $q_s = p_s M_s$ (§1.15).

1.14 Covariance with the SDF sets the risk premium [S30–S31]

Concept. Split $\mathbb{E}[\tilde{M}\tilde{R}_i]$ into a mean part and a covariance part. The mean part delivers the risk-free rate; the covariance part *is* the risk premium. An asset's premium depends only on how its return moves with $\tilde{M}$: paying off in bad times (positive covariance) is insurance and earns less than R_f; paying off in good times (negative covariance) must earn more. "Insurance pays you to hold it; risk pays you to bear it" [S31].

$$R_f = \frac{1}{\mathbb{E}[\tilde{M}]}, \qquad \mathbb{E}[\tilde{M}]\mathbb{E}[\tilde{R}_i] + \mathrm{Cov}(\tilde{M}, \tilde{R}_i) = 1 \implies \mathbb{E}[\tilde{R}_i] - R_f = -\frac{\mathrm{Cov}(\tilde{M}, \tilde{R}_i)}{\mathbb{E}[\tilde{M}]} = -R_f \,\mathrm{Cov}(\tilde{M}, \tilde{R}_i).$$

where

$R_f \mathbb{E}[\tilde{M}] = 1$	the rule for the risk-free asset; R_f is known, so it leaves the expectation [S30].
$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] + \mathrm{Cov}(X, Y)$	definition of covariance, used to split $\mathbb{E}[\tilde{M}\tilde{R}_i]$.
$-\mathrm{Cov}(\tilde{M}, \tilde{R}_i)/\mathbb{E}[\tilde{M}]$	the risk premium: its sign is the opposite of the covariance's.
two forms	identical, since $1/\mathbb{E}[\tilde{M}] = R_f$; the slides use the first.

- **Price version** *[Beyond slides]*: $P_i = \mathbb{E}[\tilde{X}_i]/R_f + \mathrm{Cov}(\tilde{M}, \tilde{X}_i)$ = risk-neutral present value + risk adjustment. Stock: $105/1.0802 + (-2.0) = 97.2 - 2.0 = 95.2$ ✓.
- **Beta form** *[Beyond slides]*: $\mathbb{E}[\tilde{R}_i] - R_f = \beta_{i,M}\,\lambda_M$, where $\beta_{i,M} = \mathrm{Cov}(\tilde{M}, \tilde{R}_i)/\mathrm{Var}(\tilde{M})$ is the asset's exposure to $\tilde{M}$ and $\lambda_M = -\mathrm{Var}(\tilde{M})/\mathbb{E}[\tilde{M}] < 0$ is the price of that exposure. Stock: $\beta = -1.95$, $\lambda = -0.0116$, premium $+2.27\%$. Insurance: $\beta = +12.3$, premium -14.3%.
- **Traps:** positive covariance with $\tilde{M}$ means a *negative* premium. Covariance with $\tilde{M}$ has the opposite sign to covariance with consumption.

△! [S31] lists gold and government bonds as insurance-like. That holds only while they pay off in bad states; their covariance with $\tilde{M}$ can change sign (e.g. bonds fell with equities in the 2022 inflation shock). The covariance, not the asset label, sets the premium.

1.15 Three consequences [Beyond slides]

(a) State prices and risk-neutral probabilities. With discrete states the price is a sum over states of "price of \$1 in that state" × payoff there.

$$P = \sum_s p_s M_s X_s = \sum_s q_s X_s, \qquad q_s = p_s M_s, \qquad \pi_s^Q = q_s R_f, \qquad P = \frac{\mathbb{E}^Q[\tilde{X}]}{R_f}.$$

where

q_s	state price: today's price of \$1 paid only in state s.
π_s^Q	risk-neutral probability: state prices rescaled to sum to 1.

Example: $q = (0.320, 0.366, 0.240)$, $\pi^Q = (0.346, 0.395, 0.259)$ vs $p = (0.3, 0.4, 0.3)$: risk-neutral probabilities *overweight bad states*.

(b) Hansen–Jagannathan bound. Because $|\mathrm{Corr}| \le 1$, no asset's Sharpe ratio can exceed the SDF's volatility relative to its mean:

$$\frac{|\mathbb{E}[\tilde{R}_i] - R_f|}{\sigma(\tilde{R}_i)} \le \frac{\sigma(\tilde{M})}{\mathbb{E}[\tilde{M}]}.$$

where

$\sigma(\tilde{M})/\mathbb{E}[\tilde{M}]$	how volatile the SDF is; high Sharpe ratios require a volatile $\tilde{M}$.

Example: bound $0.1037/0.9257 = 0.112$; the stock's Sharpe ratio is 0.112, almost on the bound because its payoff is almost perfectly negatively correlated with $\tilde{M}$. With CRRA, $\sigma(\tilde{M})/\mathbb{E}[\tilde{M}] \approx \gamma\sigma_{\Delta c}$; with consumption volatility near 2%, an equity Sharpe ratio of 0.4 needs $\gamma \approx 20$: the **equity premium puzzle** (Mehra–Prescott 1985).

(c) CAPM is a special case. If $\tilde{M}$ is linear in the market return, the covariance form becomes the CAPM:

$$\tilde{M} = a - b\tilde{R}_M \quad (b > 0) \quad \Longrightarrow \quad \mathbb{E}[\tilde{R}_i] - R_f = \beta_i\big(\mathbb{E}[\tilde{R}_M] - R_f\big), \quad \beta_i = \frac{\mathrm{Cov}(\tilde{R}_i, \tilde{R}_M)}{\mathrm{Var}(\tilde{R}_M)}.$$

where	
$b > 0$	$\tilde{M}$ is high when the market is low: market crashes are the bad states.
$\beta_i < 0$	pays when the market falls $\Rightarrow \mathrm{Cov}(\tilde{M}, \tilde{R}_i) > 0 \Rightarrow$ premium below R_f (W3's insurance result, [S31]).

Linear in K factors ($\tilde{M} = a - b^\top \tilde{f}$) gives the W4 multi-factor model. "Which asset pricing model?" = "which $\tilde{M}$?".

1.16 Implementation [Beyond slides] (tested)

```
import numpy as np
p = np.array([0.3, 0.4, 0.3]) # recession, normal, boom
C0, delta = 100.0, 0.96
C1 = np.array([90.0, 105.0, 120.0])
M = delta * C0 / C1 # log utility: M = delta * C0 / C1
E = lambda x: float(p @ x)
Rf = 1 / E(M)
print(np.round(M, 4), round(E(M), 4), round(Rf, 4)) # [1.0667 0.9143 0.8] 0.9257 1.0802
for name, X in [("bond", np.ones(3)), ("insurance", np.array([1.0, 0, 0])),
            ("stock", np.array([80.0, 105, 130]))]:
   P = E(M * X); R = X / P
   cov = E(M * R) - E(M) * E(R)
   print(name, round(P, 4), round(E(R), 4), # price, expected return
        round(E(R) - Rf, 4), round(-cov / E(M), 4)) # premium = -Cov(M,R)/E[M]
# bond 0.9257 1.0802 0.0 0.0 | insurance 0.32 0.9375 -0.1427 -0.1427 | stock 95.2 1.1029 0.0227 0.0227
Re = np.array([80.0, 105, 130]) / 95.2 - Rf # excess return of the stock
print(round(E(M * Re), 10)) # 0.0: zero-cost position
print(round(1 / C0, 4), round(delta * 0.3 * (1 / 0.32) / 90, 4)) # Euler: 0.01 0.01
```

2 Comparison tables

2.1 The two first-order conditions [S17, S20]

	FOC in C_0 (Euler)	FOC in w_i (allocation)
Question	eat or invest the marginal dollar?	which asset gets the marginal dollar?
Perturbation	move \$1 between today and the portfolio	move weight from asset j to asset i
Key derivative	$\partial\tilde{C}_1/\partial C_0 = -\tilde{R}_p$	$\partial\tilde{C}_1/\partial w_i = (W_0 - C_0)\tilde{R}_i$
Condition	$U'(C_0^*) = \delta\mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_p]$	$\mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_i]$ equal for all i
Picture	level balance beam [S19]	one U' schedule shared by all assets [S21]
Pins down	how much to save	relative pricing of assets
Examinable	result; derivation "(optional)"	result and derivation

2.2 One equation, many forms

Form	Equation	Slides	Best for
Return form	$\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$	[S23]	testing with return data
Price form	$P_i = \mathbb{E}[\tilde{M}\tilde{X}_i]$	[S25], [S27]	valuing any payoff
Excess-return form	$\mathbb{E}[\tilde{M}\tilde{R}_i^e] = 0$	[S26]	zero-cost positions
Risk-free rate	$R_f = 1/\mathbb{E}[\tilde{M}]$	[S30]	short rate from preferences
Covariance form	$\mathbb{E}[\tilde{R}_i] - R_f = -\operatorname{Cov}(\tilde{M}, \tilde{R}_i)/\mathbb{E}[\tilde{M}]$	[S30]–[S31]	explaining risk premia
State-price form	$P_i = \sum_s q_s X_{i,s}$	*[Beyond slides]*	discrete-state problems
Risk-neutral form	$P_i = \mathbb{E}^Q[\tilde{X}_i]/R_f$	*[Beyond slides]*	derivatives pricing

2.3 Insurance vs procyclical assets [S29, S31]

	Insurance-like	Procyclical	Independent of $\tilde{M}$
Pays off	when times are bad	when times are good	regardless of state
Examples	gold, government bonds, puts, insurance [S31]	equities, cyclical firms, leverage, the market [S31]	idiosyncratic bets (coin flip)
$\operatorname{Cov}(\tilde{M}, \tilde{R})$	> 0	< 0	0
Price	high ("dear")	low ("cheap")	$\mathbb{E}[\tilde{X}]/R_f$
Expected return	$< R_f$	$> R_f$	$= R_f$
W3 analogue	$\beta < 0$	$\beta > 0$	$\beta = 0$

2.4 Certain vs stochastic discounting [S28]

	Certain payoff	Random payoff
Price	$P = mX$	$P = \mathbb{E}[\tilde{M}\tilde{X}]$
Discount factor	one number, $m = 1/(1+r)$	one value per state, $\tilde{M}$
Depends on	time only	time (δ) and state ($U'(\tilde{C}_1)/U'(C_0)$)
Link	$m = 1/R_f = \mathbb{E}[\tilde{M}]$ *[Beyond slides]*	

2.5 SDF vs CAPM (Week 3)

	CAPM (W3)	SDF (L6)
Starting point	mean–variance investors + market clearing	any expected-utility investor's first-order conditions
Bad state means	low market return	low consumption (high U')
What prices risk	$\beta_i = \operatorname{Cov}(\tilde{R}_i, \tilde{R}_M)/\operatorname{Var}(\tilde{R}_M)$	$\operatorname{Cov}(\tilde{M}, \tilde{R}_i)$
Pricing equation	$\mathbb{E}[\tilde{R}_i] - R_f = \beta_i(\mathbb{E}[\tilde{R}_M] - R_f)$	$\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$
Relation	special case with $\tilde{M} = a - b\tilde{R}_M$	general framework

3 Master formula sheet

Each line repeats the key symbols so the sheet can be used on its own.

Budget and preferences [S3, S6–S9, S18]

- $C_0 + S = W_0$; $\sum_i w_i = 1$; $\tilde{C}_1 = (W_0 - C_0)\sum_i w_i\tilde{R}_i$; $U' > 0$, $U'' < 0$; objective $U(C_0) + \delta\mathbb{E}[U(\tilde{C}_1)]$.
  where $\tilde{R}$ = gross return $(1 + \tilde{r})$; w_i = share of the invested pot; δ = patience (discounts utility).

Problem and first-order conditions [S15, S17, S20–S21]

- $\mathcal{L} = U(C_0) + \delta\mathbb{E}[U(\tilde{C}_1)] + \lambda(1 - \sum_i w_i)$.
- Euler: $U'(C_0^*) = \delta\mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_p]$; allocation: $\delta(W_0 - C_0)\mathbb{E}[U'(\tilde{C}_1)\tilde{R}_i] = \lambda$ for all i.
  where $\partial\tilde{C}_1/\partial C_0 = -\tilde{R}_p$; $\partial\tilde{C}_1/\partial w_i = (W_0 - C_0)\tilde{R}_i$; $\lambda = (W_0 - C_0)U'(C_0^*)$ *[Beyond slides]*.
- Per asset: $U'(C_0^*) = \delta\mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_i]$.

SDF and pricing [S23–S31]

- $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$; $\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$; $P_i = \mathbb{E}[\tilde{M}\tilde{X}_i]$; $\mathbb{E}[\tilde{M}\tilde{R}_i^e] = 0$.
  where $\tilde{R}_i = \tilde{X}_i/P_i$; $\tilde{R}_i^e = \tilde{R}_i - R_f$; $\tilde{M}$ large in bad states.
- $R_f = 1/\mathbb{E}[\tilde{M}]$; $\mathbb{E}[\tilde{R}_i] - R_f = -\operatorname{Cov}(\tilde{M}, \tilde{R}_i)/\mathbb{E}[\tilde{M}] = -R_f \operatorname{Cov}(\tilde{M}, \tilde{R}_i)$.
  where $\operatorname{Cov} > 0$ (insurance) $\Rightarrow$ premium < 0; $\operatorname{Cov} < 0$ (procyclical) $\Rightarrow$ premium > 0.
- Certain payoff: $P = mX$, $m = 1/(1+r) = \mathbb{E}[\tilde{M}]$.

Special utilities *[Beyond slides]*

- Log: $\tilde{M} = \delta C_0/\tilde{C}_1$, $C_0^* = W_0/(1+\delta)$; CRRA: $\tilde{M} = \delta(\tilde{C}_1/C_0)^{-\gamma}$; quadratic: $\tilde{M}$ linear in $\tilde{C}_1$.

Further consequences *[Beyond slides]*

- $P_i = \mathbb{E}[\tilde{X}_i]/R_f + \operatorname{Cov}(\tilde{M}, \tilde{X}_i)$; $q_s = p_s M_s$, $\pi_s^Q = q_s R_f$, $P = \mathbb{E}^Q[\tilde{X}]/R_f$.
- HJ: $|\mathbb{E}[\tilde{R}] - R_f|/\sigma(\tilde{R}) \le \sigma(\tilde{M})/\mathbb{E}[\tilde{M}]$; $\tilde{M} = a - b\tilde{R}_M \Rightarrow$ CAPM.

4 Glossary

- **Allocation rule:** FOC in w_i; $\mathbb{E}[U'(\tilde{C}_1)\tilde{R}_i]$ is equal across assets [S20].
- **Consumption smoothing:** spreading consumption evenly over time and states; optimal because U is concave [S6].
- **Euler equation:** FOC in C_0; marginal utility today = discounted expected marginal utility of investing the dollar [S17, S19].
- **Excess return:** $\tilde{R}_i - R_f$, the return on a zero-cost position; $\mathbb{E}[\tilde{M}\tilde{R}_i^e] = 0$ [S26].
- **Gross return:** what \$1 becomes, $1 + r$ [S8].
- **Hansen–Jagannathan bound:** Sharpe ratio $\le \sigma(\tilde{M})/\mathbb{E}[\tilde{M}]$ *[Beyond slides]*.
- **Insurance-like asset:** pays in bad states; $\operatorname{Cov}(\tilde{M}, \tilde{R}) > 0$; negative premium [S31].
- **IMRS:** intertemporal marginal rate of substitution; how many of today's dollars a dollar in a future state is worth; equals $\tilde{M}$ [S24].
- **Lagrange multiplier λ:** shadow value of $\sum w_i = 1$ [S15, S20].
- **Marginal utility U':** extra utility from one more dollar; high when consumption is low [S18].
- **Patience δ:** utility discount factor [S9].
- **Pricing kernel:** another name for the SDF [S24].
- **Procyclical asset:** pays in good states; $\operatorname{Cov}(\tilde{M}, \tilde{R}) < 0$; positive premium [S31].
- **State price q_s:** price today of \$1 paid only in state s; $q_s = p_s M_s$ *[Beyond slides]*.
- **Stochastic discount factor $\tilde{M}$:** $\delta U'(\tilde{C}_1)/U'(C_0)$; a random discount factor, one value per state [S24, S28].
- **Time-separable utility:** today's and tomorrow's utility add with no interaction *[Beyond slides]*.

5 Self-test

Questions

Q1 ([S3, S19]). Explain why, at the optimum, a dollar is worth the same eaten or invested. What should the investor do if the invested side of the balance is heavier?

Q2 ([S18, S29]). Why must future payoffs be weighted by U' rather than by probability alone? Why is $\tilde{M}$ high in recessions?

Q3 ([S17], optional). Derive the Euler equation. Where does the minus sign come from, and why can't $U'(\tilde{C}_1)$ be taken outside the expectation?

Q4 ([S20–S21]). Derive the allocation rule and explain why the Euler equation then holds asset by asset.

Q5 ([S25, S30]). Two equally likely states with $\tilde{M} = (1.2,\ 0.7)$ in (bad, good). Price a riskless bond paying \$1 and find R_f. Price asset A with payoff (50, 150) and asset B with payoff (150, 50); find their expected returns and risk premia.

Q6 ([S26]). Show $\mathbb{E}[\tilde{M}\tilde{R}_i^e] = 0$. Does it imply $\mathbb{E}[\tilde{R}_i^e] = 0$?

Q7 ([S30]). Derive the covariance form in four steps and use it to verify the premia in Q5.

Q8 (*[Beyond slides]*). $U = \ln C$, $\delta = 0.96$, $W_0 = 196$. Find C_0^* and S^*. Date-1 consumption is 90, 105 or 120 with probabilities 0.3, 0.4, 0.3: find $\tilde{M}$, R_f and the price of a claim paying \$1 only in the recession state.

Q9 (*[Beyond slides]*). Same states, CRRA with $\gamma = 2$ ($\delta = 0.96$, $C_0 = 100$). Find $\tilde{M}$ and R_f, price the payoff (80, 105, 130) and find its premium. Compare with log utility.

Q10 (traps). True or false? (a) Higher return variance means a higher expected return. (b) $P = \mathbb{E}[\tilde{M}]\mathbb{E}[\tilde{X}]$. (c) With net returns, $\mathbb{E}[\tilde{M}\tilde{r}] = 1$. (d) δ discounts dollars. (e) λ measures risk aversion. (f) Government bonds always have a negative premium.

Solutions

S1. If eating or investing the last dollar gave more utility, moving a dollar that way would raise V, so the plan could not be optimal. Invested side heavier: consume less now and invest more until the beam is level.

S2. $U'' < 0$ makes a dollar worth more when consumption is low, so equal-probability payoffs in different states are not equally valuable. In a recession $\tilde{C}_1$ is low, $U'(\tilde{C}_1)$ is high, so $\tilde{M} = \delta U'(\tilde{C}_1)/U'(C_0)$ is high.

S3. $dV/dC_0 = U'(C_0) + \delta\mathbb{E}[U'(\tilde{C}_1)\,\partial\tilde{C}_1/\partial C_0] = 0$ with $\partial\tilde{C}_1/\partial C_0 = -\tilde{R}_p$ (a dollar eaten is a dollar not invested) $\Rightarrow U'(C_0^*) = \delta\mathbb{E}[U'(\tilde{C}_1^*)\tilde{R}_p]$. $U'(\tilde{C}_1)$ is random and depends on $\tilde{R}_p$, so $\mathbb{E}[U'\tilde{R}_p] \neq \mathbb{E}[U']\mathbb{E}[\tilde{R}_p]$.

S4. $L = \delta\mathbb{E}[U(\tilde{C}_1)] - \lambda(\sum_j w_j - 1)$; $\partial/\partial w_i$: $\delta(W_0 - C_0)\mathbb{E}[U'\tilde{R}_i] = \lambda$, the same for all i. The Euler term $\mathbb{E}[U'\tilde{R}_p] = \sum_i w_i\mathbb{E}[U'\tilde{R}_i]$ is an average of equal numbers, so it equals each: $U'(C_0) = \delta\mathbb{E}[U'\tilde{R}_i]$ for every i.

S5. Bond $= \mathbb{E}[\tilde{M}] = \mathbf{0.95}$, $R_f = \mathbf{1.0526}$. A: $0.5(1.2)(50) + 0.5(0.7)(150) = \mathbf{82.5}$, $\mathbb{E}[\tilde{R}] = 100/82.5 = 1.2121$, premium $+\mathbf{15.9}\%$. B: $\mathbf{107.5}$, $\mathbb{E}[\tilde{R}] = 0.9302$, premium $-\mathbf{12.2}\%$ (B is insurance).

S6. $\mathbb{E}[\tilde{M}\tilde{R}_i] = 1 = \mathbb{E}[\tilde{M}R_f]$; subtract: $\mathbb{E}[\tilde{M}(\tilde{R}_i - R_f)] = 0$, because a zero-cost position has price zero. No: $\mathbb{E}[\tilde{R}_i^e] = -\operatorname{Cov}(\tilde{M}, \tilde{R}_i)/\mathbb{E}[\tilde{M}]$, zero only if the covariance is zero.

S7. (1) $\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$; (2) $R_f\mathbb{E}[\tilde{M}] = 1$; (3) $\mathbb{E}[\tilde{M}]\mathbb{E}[\tilde{R}_i] + \operatorname{Cov} = 1$; (4) divide by $\mathbb{E}[\tilde{M}]$. A: $\tilde{R}_A = (0.606, 1.818)$, $\operatorname{Cov}(\tilde{M}, \tilde{R}_A) = 1 - 0.95 \times 1.2121 = -0.1515$, premium $0.1515/0.95 = +15.9\%$ ✓. B: Cov $= +0.1163$, premium -12.2% ✓.

S8. $1/C_0 = \delta/(W_0 - C_0) \Rightarrow C_0^* = 196/1.96 = \mathbf{100}$, $S^* = \mathbf{96}$. $\tilde{M} = 0.96 \times 100/C_1 = (\mathbf{1.0667},\ \mathbf{0.9143},\ \mathbf{0.8000})$, $\mathbb{E}[\tilde{M}] = 0.9257$, $R_f = \mathbf{1.0802}$. Claim: $0.3 \times 1.0667 = \mathbf{0.320}$, $\mathbb{E}[\tilde{R}] = 0.9375$ (premium -14.3%).

S9. $\tilde{M} = 0.96(C_1/100)^{-2} = (\mathbf{1.1852},\ \mathbf{0.8707},\ \mathbf{0.6667})$, $\mathbb{E}[\tilde{M}] = 0.9039$, $R_f = \mathbf{1.1064}$. $P = 28.44{+}36.57{+}26.00 = \mathbf{91.02}$, $\mathbb{E}[\tilde{R}] = 1.1536$, premium $+\mathbf{4.73}\%$. Both rise vs log ($+2.27\%$, 1.0802): more curvature makes $\tilde{M}$ more dispersed (bigger premium) and strengthens the wish to borrow against expected growth (higher R_f).

S10. (a) False: only $\operatorname{Cov}(\tilde{M}, \tilde{R})$ is priced (coin flip earns R_f). (b) False: $+\operatorname{Cov}(\tilde{M}, \tilde{X})$. (c) False: $\mathbb{E}[\tilde{M}\tilde{r}] = 1 - \mathbb{E}[\tilde{M}]$. (d) False: δ discounts utility; $\tilde{M}$ discounts dollars. (e) False: shadow value of $\sum w_i = 1$. (f) False: only while they pay in bad states.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title: Stochastic Discount Factor	header
S2	"How do you spend your salary?"	1.1
S3	Spend (C_0) or invest ($S = W_0 - C_0$)	1.1
S4	Uncertainty; tilde = random	1.2
S5	"A dollar is worth more when you are poor" △!	1.3
S6	Diminishing marginal utility; smooth consumption	1.3
S7	One period, two decisions; symbol table	1.4
S8	Budget conditions; gross returns	1.4
S9	Objective $U(C_0) + \delta\mathbb{E}[U(\tilde{C}_1)]$	1.5
S10	Recap: four points	1.5
S11	How wealth evolves; total utility △!	1.2, 1.5
S12	Where we are: which choices are ours?	1.6
S13	Two decisions: we choose vs the market decides	1.4, 1.6
S14	The investment leg	1.4
S15	Problem, Lagrangian, two FOCs	1.6
S16	In-class quiz Q1–3	exam context
S17	FOC in C_0: Euler equation (optional)	1.7
S18	$U' > 0$, $U'' < 0$; weight states by U'	1.3
S19	Euler as a balance beam	1.7
S20	FOC in w_i: allocation rule	1.8
S21	One objective, two conditions △!	1.9
S22	In-class quiz Q4–6	exam context
S23	$\mathbb{E}[\tilde{M}\tilde{R}_i] = 1$ (derivation not tested)	1.9
S24	SDF: pieces, one value per state, three names △!	1.10
S25	From returns to prices	1.11
S26	Excess-return form	1.12
S27	Today's price is the expected payoff	1.11
S28	Why "stochastic discount factor"; $P = mX \to P = \mathbb{E}[\tilde{M}\tilde{X}]$	1.10
S29	SDF largest when a dollar is scarce △!	1.13
S30	Covariance sets the risk premium; $R_f = 1/\mathbb{E}[\tilde{M}]$	1.14
S31	Insurance vs procyclical assets △!	1.14, 2.3
S32	In-class quiz Q7–9	exam context

Appendix B. Flag summary (△!)

1. **[S17], [S23]** Marked "(optional)" and "(derivation not tested)": learn the results.
2. **[S5]** "worth more you are poor": missing "when".
3. **[S11]** Uses the net return ($1 + \tilde{r}$); all other slides use the gross $\tilde{R} = 1 + \tilde{r}$.
4. **[S15] vs [S20]** $+\lambda(1 - \sum w_i)$ and $-\lambda(\sum w_j - 1)$ are identical.
5. **[S20]** λ is a Lagrange multiplier, not risk aversion or a factor premium.
6. **[S21]** $\lambda/\delta(W_0 - C_0)$ means $\lambda/[\delta(W_0 - C_0)]$.
7. **[S20]–[S23]** "Every asset" needs interior weights (no binding short-sale limits).
8. **[S24]** An IMRS is in today's *dollars*, not "happiness".
9. **[S25], [S27]** "Pound" and "dollar" used interchangeably.
10. **[S29]** $\tilde{M}$ is the value of \$1 in a state *per unit probability*; the state price is $q_s = p_s M_s$.
11. **[S31]** Gold and government bonds are insurance-like only while they pay in bad states.